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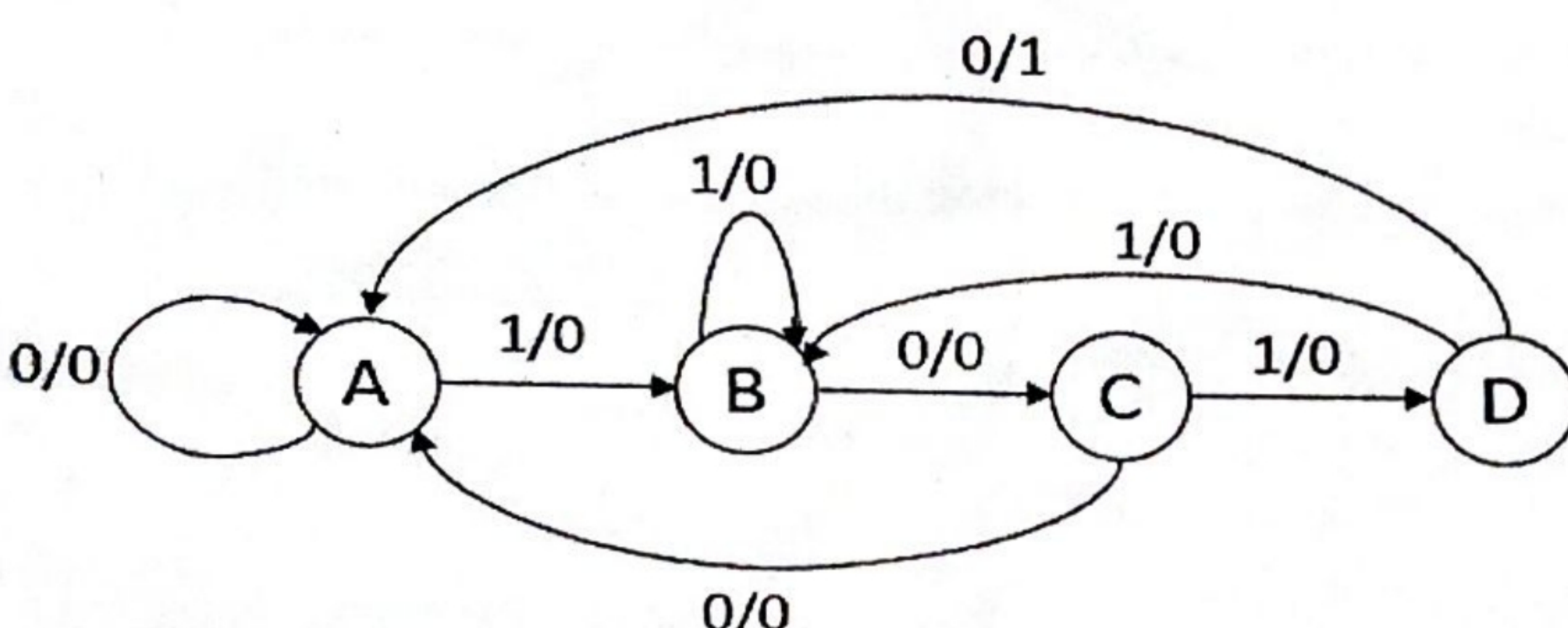
PAPER CODE: BET-C306
SEMESTER EXAMINATION- DECEMBER 2024
CLASS: B.TECH SEMESTER: III
SUBJECT: DIGITAL SYSTEM DESIGN
PAPER TITLE: DIGITAL SYSTEM DESIGN

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: A and B. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>five</i> questions in about 150 words each. Each question carries six marks. (5 X 6 = 30 Marks)			
Q-1	How do perform subtraction using the 2's compliment method, Explain with an example.	CO1	L1
Q-2	(i) Convert $(542)_{10}$ to $(...)_{2}$. (ii) Convert $(547)_8$ to $(...)_{\text{gray}}$	CO1	L2
Q-3	(i) Add 587.40 to 254.60 in BCD code. (ii) Subtract 543.51 from 652.12 in XS-3 code.	CO1	L2
Q-4	Design a full adder circuit using basic logic gates. Provide the truth table for the full adder.	CO2	L3
Q-5	How can X-OR gate be used as an inverter? Implement X-OR gate using NAND gate.	CO1	L1
Q-6	What is decoder? Realize 4X16 decoder using two 3X8 decoder.	CO2	L4
Q-7	Apply Demorgan's theorem to the expression : $F = \overline{AB} (CD + \overline{EF})(\overline{AB} + \overline{CD})$	CO1	L3
Q-8	Discuss the different types of flip-flops. Differentiate combinational and sequential circuits.	CO3	L2
Q-9	Design and explain parallel in, parallel out shift register.	CO3	L2
Q-10	Explain the concept of fan-in, fan-out and noise margin in digital logic families.	CO1	L1
SECTION-B (LONG ANSWER TYPE QUESTIONS)		CO	BL
Instructions: Answer any <i>four</i> questions in detail. Each question carries 10 marks. (4 X 10 = Marks)			
Q-1	Simplify the following function using 4 variable k-map: (i) $F = \sum m(0, 2, 6, 10, 11, 12, 13) + d(3, 4, 5, 14, 15)$ (ii) $F = ABD + \overline{A}\overline{C}\overline{D} + \overline{A}B + \overline{A}\overline{C}\overline{D} + \overline{A}\overline{B}D$	CO1	L2
Q-2	Reduce following expression using tabular method and implement reduced function using NAND logic. $F = (A, B, C, D) = \sum m(2, 3, 8, 12, 13) + d(10, 14)$	CO1	L2

Q-3	Realize the logic expression using a (i) 8:1 MUX (ii) 4 line to 16 line decoder $F(A,B,C,D)=\sum(0,1,3,5,8,11,12,14,15)$	CO2	L4
Q-4	What is Race-round condition in flip flop; explain it and how to remove it.	CO3	L1
Q-5	Convert : (i) JK to SR flip flop (ii) SR to D flip flop	CO3	L3
Q-6	Design a circuit using D flip flop that will function as per the state diagram shown below  <pre> graph LR A((A)) -- "0/0" --> A A -- "1/0" --> B((B)) B -- "1/0" --> B B -- "0/0" --> C((C)) C -- "1/0" --> D((D)) D -- "1/0" --> B D -- "0/1" --> A </pre>	CO3	L6
Q-7	Design a Modulo-9 synchronous counter using T Flip Flops.	CO3	L5
Q-8	Draw and explain the circuit diagram for Integrated Injection Logic (I ² L) NOR gate and CMOS NAND gate.	CO1	L2